Coupling networks acc. EN 61000-4-16

Coupling networks for powerline conductors

For each wire the coupling network for powerline conductors is made of a serious connection of a resistor and a capacitor. Coupling networks of each wire are connected to establish the coupling network of the corresponding M-type.

Value of the capacitor is C = 1,0 µF.

Value of the resistor is R = 100 x n Ω, where n is the number of the wires ($n \geq 2$).

Values of capacitor and resistor shall match with a limiting deviation of 1 %.

For direct current tests the 1,0 µF capacitors shall be short circuited.

For safety reasons coupling networks M2 and M3 are separated units for DC tests and AC tests. Short circuiting the capacitor by mistake while an alternating current is applied inevitably destroys the coupling network.

Each connection which is not under test must be grounded (SW2). For this reason an isolated BNC jumper plug is included.

Coupling networks for communication lines

For balanced communication lines and similar lines a "T" network is used.

The "T" network is made of capacitors (C = 4,7 µF), resistors (R = 200 Ω) and inductances (L = 2 x 38 mH / bifilar winding).

All components shall match with a limiting deviation to ensure that there is no significant reduction of the differential to common mode conversion loss of the EUT .

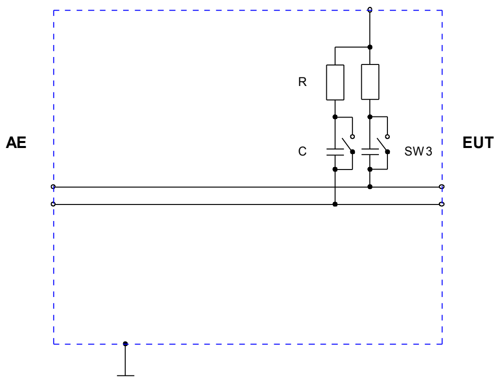
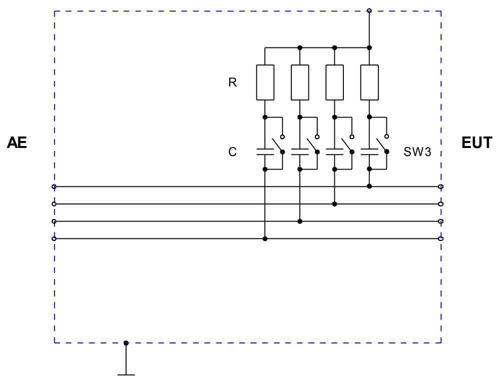
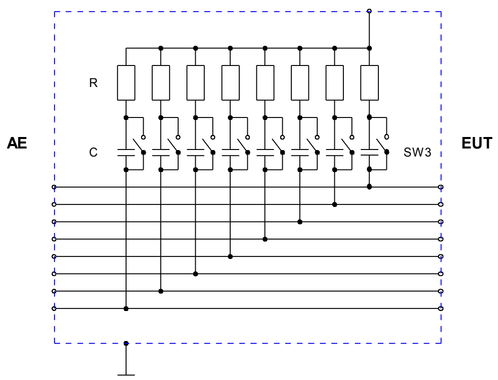
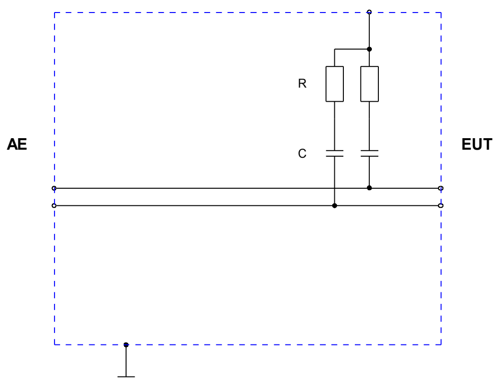
For direct current tests the 1,0 µF capacitors shall be short circuited.

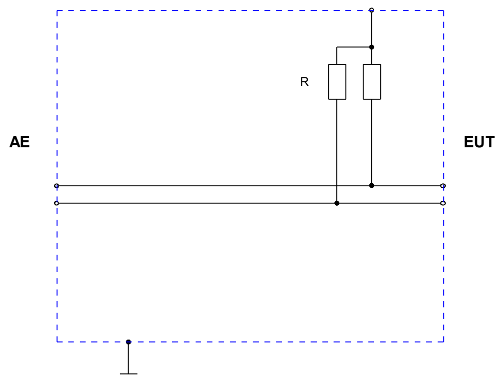
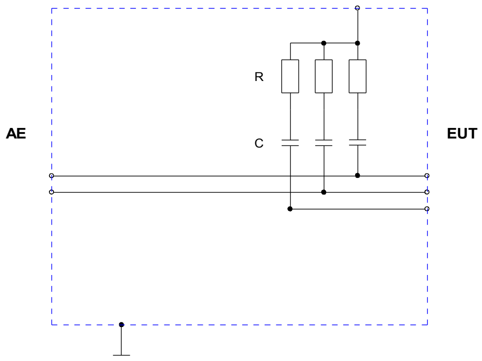
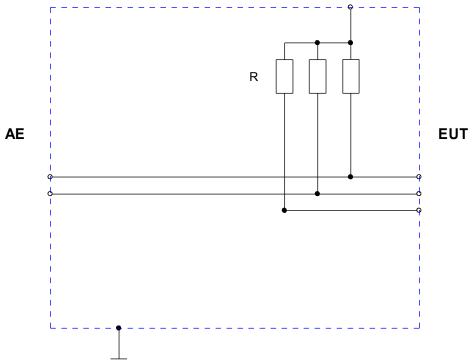
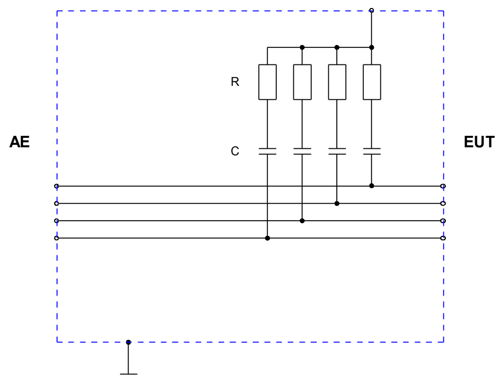
Each connection which is not under test must be grounded (SW2). For this reason an isolated BNC jumper plug is included.

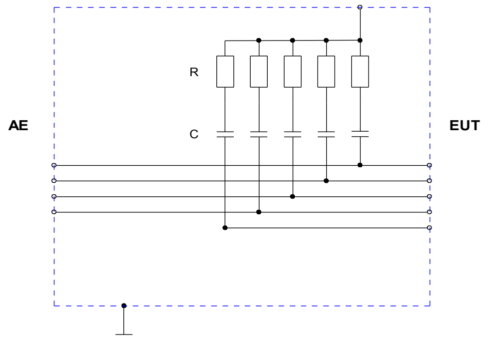
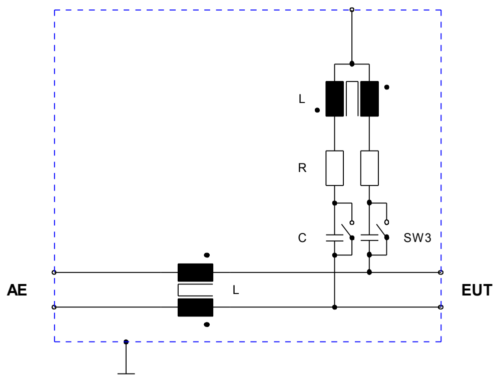
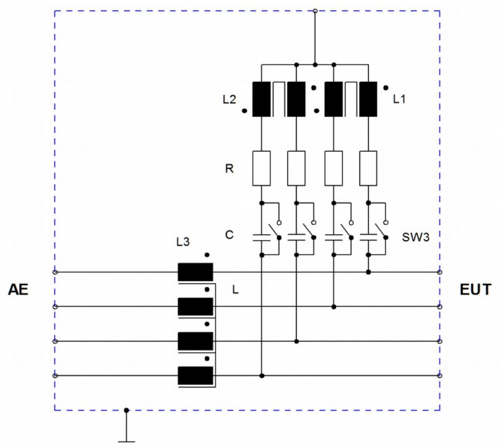
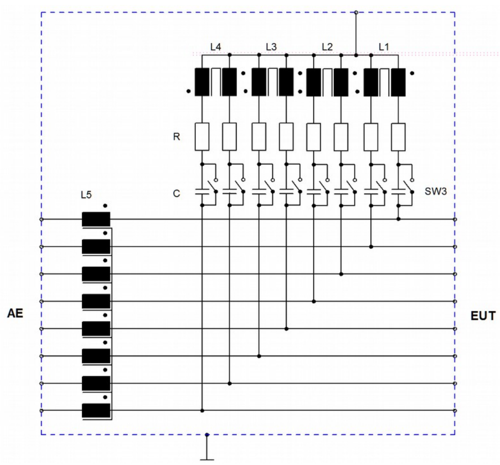
Important safety instructions for immunity tests according to EN 61000-4-16

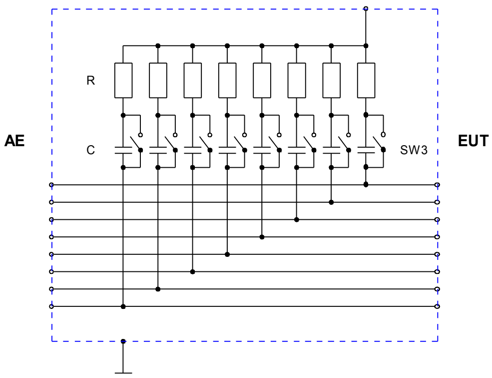
Please, notice the following safety instructions for immunity tests on AC and DC supply lines (AC > 30 V / DC > 60 V):

- The coupling network must be in tight connection to the ground reference plane!
- An additional cable connection between coupling network baseplate (threaded bolts) and ground reference plane is recommended!
- Always ground the coupling network before connecting any power line to the AE-port!
- Never open the ground connection between coupling network and ground reference plane before removing the supply lines from the AE-port!

Type	*Simplified diagram*	*Description*
AF2		Coupling network acc. EN 61000-4-16 for unscreened, unbalanced lines Frequency range: DC / 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rocker switch.
AF4		Coupling network acc. EN 61000-4-16 for unscreened, unbalanced lines Test level: 50 V cont.; 300 V (1sec) power frequency Test level: 50 V cont. EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rotary switch.
AF8		Coupling network acc. EN 61000-4-16 for unscreened, unbalanced lines Frequency range: DC / 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rotary switch.
M2/AC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Frequency range: 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - Connector: 250 VAC / 32 A (50 VDC / 32 A) Connector EUT / AE: 4mm safety banana jack For AC tests only

Type	Simplified diagram	Description
M2/DC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Frequency range: DC Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 32 A Connector EUT / AE: 4mm safety banana jack For DC tests only!
M3/AC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Use for EUT with functional earth! Frequency range: 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - Connector: 250 VAC / 32 A (50 VDC / 32 A) Connector EUT / AE: 4mm safety banana jack For AC tests only
M3/DC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Use for EUT with functional earth! Frequency range: DC Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 32 A Connector EUT / AE: 4mm safety banana jack For DC tests only!
M4/AC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Frequency range: 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - Connector: 250 VAC / 32 A Connector EUT / AE: 4mm safety banana jack For AC tests only

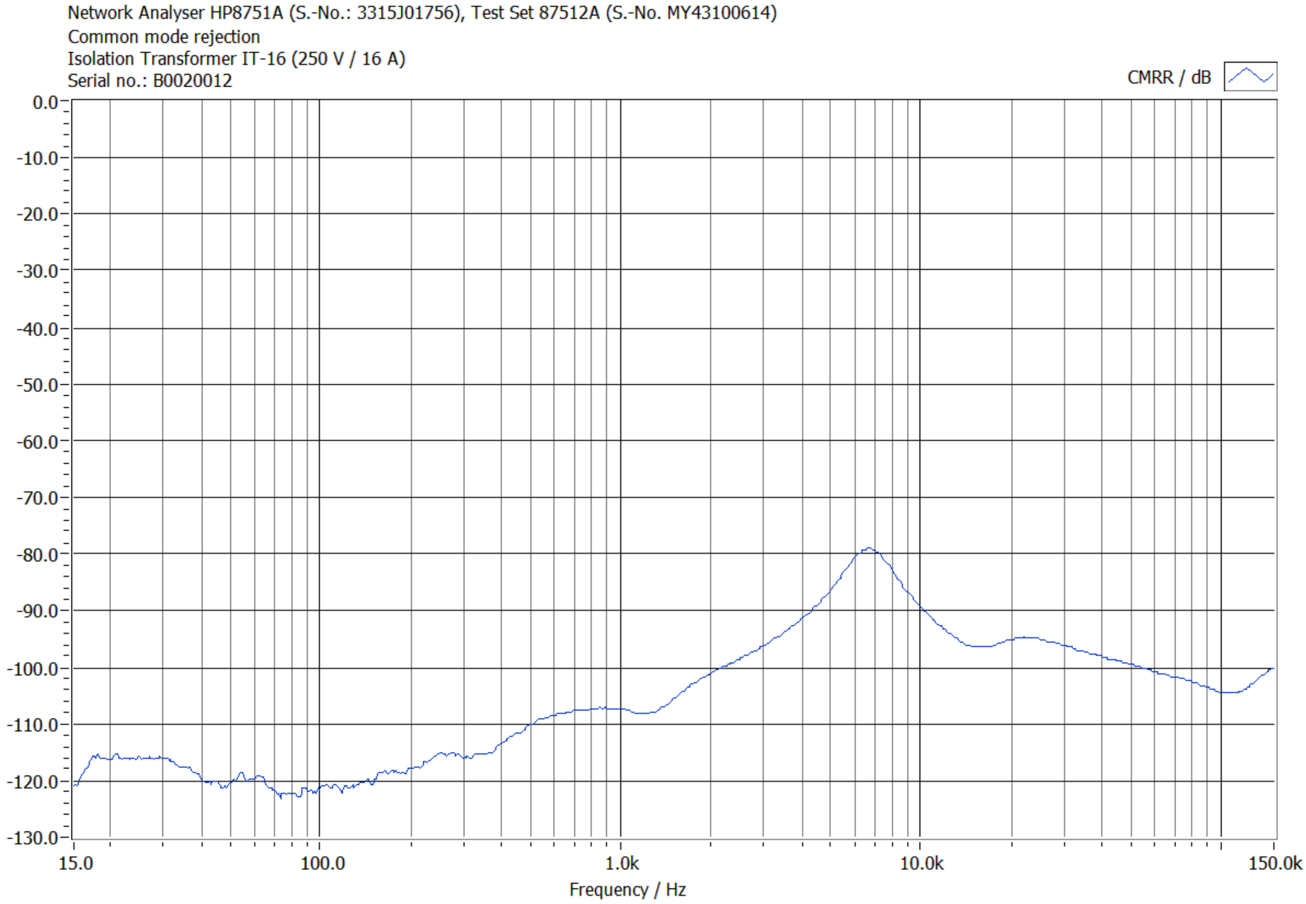
Type	*Simplified diagram*	*Description*
M5/AC		Coupling network acc. EN 61000-4-16 for unscreened powerline conductors Use for EUT with functional earth! Frequency range: 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - Connector: 250 VAC / 32 A Connector EUT / AE: 4mm safety banana jack <u>For AC tests only</u>
T2		Coupling network acc. EN 61000-4-16 for unscreened, balanced lines Frequency range: DC / 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rocker switch. Differential to common mode conversion loss (15 Hz to 150 kHz): 60 dB Insulation: > 1 kV (50/60 Hz)
T4		Coupling network acc. EN 61000-4-16 for unscreened, balanced lines Frequency range: DC / 15 Hz to 150 kHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rotary switch. Differential to common mode conversion loss (15 Hz to 150 kHz): 60 dB Insulation: 1 kV, 50/60 Hz, 1 min
T8		Coupling network acc. EN 61000-4-16 for unscreened, balanced lines Frequency range: DC / 15 Hz to 150 kHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rotary switch. Differential to common mode conversion loss (15 Hz to 150 kHz): 60 dB Insulation: 1 kV, 50/60 Hz, 1 min

Type	*Simplified diagram*	*Description*
RJ45		Coupling network acc. EN 61000-4-16 for unscreened, balanced lines Frequency range: DC / 15 Hz to 150 KHz Test level: 50 V cont.; 300 V (1sec) power frequency EUT / AE - port: 50 V / 0,5 A Connector EUT / AE: clamp terminal For DC tests the capacitors are short circuited by a rotary switch.

IT-6 / IT-16 / IT-20

Isolations transformer 1380VA / 3680VA / 4600VA (EN 61000-4-16)

Common mode rejection (typical data):

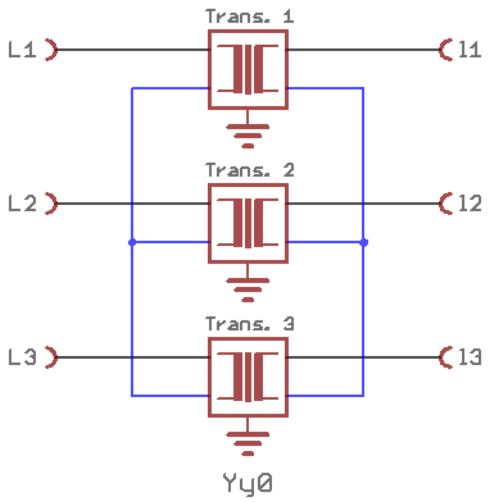


Specifications	IT-6	IT-16	IT-20
Prim	230 V	230V	230V
Sec	230 V / 6 A	230 V / 16 A	230 V / 20 A
Differential to common mode conversion loss (15 Hz to 150 kHz)	60 dB	60 dB	60 dB
Insulation	> 1 kV (50/60 Hz)	> 1 kV (50/60 Hz)	> 1 kV (50/60 Hz)
Dimension (W / D / H)	330 mm x 230 mm x 111 mm	400 mm x 310 mm x 181 mm	400 mm x 310 mm x 181 mm
Weight	~18 kg	~34 kg	~45 kg

IT-16/20 in a balanced three-phase system

In a balanced three-phase power supply system, three conductors each carry an alternating current of the same frequency and voltage relative to a common reference but with a phase difference of one third the period. Such a system may be galvanically isolated with three identical isolation transformers with high common mode rejection.

A three-phase system constructed by three individual single phase transformers requires a balanced load, i.e. the neutral wire does not carry any current!



The primary and secondary windings of the three IT- transformer are connected in star-configuration.

For safety reasons the (blue) neutral wire should not be connected to the EUT!

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